Thermodynamics and Statistical Physics

10th Exercise Class

Phases and phase changes; Clausius–Clapeyron relation

Exercises for the Class

1. A calorimeter with negligible heat capacity contains 1 kg of ice with a temperature of -36 °C. Water vapor with temperature of 100 °C is very slowly injected into the calorimeter. Depending on the mass of vapor find the possible phases in the calorimeter when the thermal interaction ends.

   Use the following data: Latent heat of evaporation: 2260 kJ/kg, latent heat of fusion: 334 kJ/kg, specific heat of water and ice are 4180 J/(kg K), 2090 J/(kg K), respectively.

2. The volume of water vapor at 100 °C and 50 kPa pressure is reduced at constant temperature. What is the pressure of the vapor when its volume is only a quarter of its original volume? What is the ratio of water to water vapor in the system at this moment?

3. A small object with negligible heat capacity is dropped into supercooled water at -20 °C. What fraction of the water will freeze?

4. Determine how the pressure of a saturated water vapor depends on temperature, if we know that at $T_0 = 100$ °C its pressure is $p_0 = 100$ kPa. Assume that the latent heat of evaporation L does not depend on the temperature in the range around T_0, and assume that the volume of the vapor is much larger than the volume of water, and assume that the vapor is an ideal gas.

5. Determine the heat of vaporisation of water at atmospheric pressure as a function of temperature between 0 °C and 100 °C. We can use the specific heat of water and water vapor at constant pressure, the known value of the latent heat of evaporation at 100 °C.

6. The vapor pressure of water is 94.4 kPa at 98 °C and 108.9 kPa at 102 °C. The density of liquid water at 100 °C is 958 kg/m^3. What is the density of steam at that temperature?

Homework to Be Submitted

1. Consider an air parcel in the atmosphere which contains saturated vapor and air. This air parcel rises adiabatically starting from a temperature $T_\mathrm{i} = 17$ °C and pressure $p_\mathrm{i} = 10^5$ Pa. The mass mixing ratio of water vapor (the mass of water vapor relative to the total mass) is $\phi = 0.01$.

   Find the temperature T at which liquid water starts forming in the air parcel. Assume that the water content in the air parcel stays constant during the rise. Use $L = 2460$ kJ/kg and the saturated water pressure at 17 °C is 1940 Pa. Consider the vapor as a triatomic ideal gas with 6 degrees of freedom.

Homework for Practice

1. A chunk of ice of mass $m_1 = 100$ g at a temperature $T_1 = 0$ °C was placed in a calorimeter in which water of mass $m_2 = 100$ g was at a temperature T_2. Assuming the heat capacity of the calorimeter to be negligible, find the entropy increment of the system by the moment the thermal equilibrium is reached. Consider two cases: (a) $T_2 = 60$ °C; (b) $T_2 = 94$ °C.

2. Assuming the saturated water vapour to be ideal, find its pressure at the temperature 101.1 °C.

3. Water of mass $m = 20$ g is enclosed in a thermally insulated cylinder at the temperature of $T_0 = 0$ °C under a weightless piston whose area is $A = 410$ cm^2. The outside pressure is equal to standard atmospheric pressure. To what height will the piston rise when the water absorbs $Q = 20$ kJ of heat?

4. An ice which was initially under standard conditions was compressed up to the pressure $p = = 640$ atm. Assuming the lowering of the ice melting temperature to be a linear function of pressure under the given conditions, find what fraction of the ice melted. The specific volume of water is less than that of ice by $\Delta V' = = 0.09$ cm^3/g.

5. How much heat must be removed from 2 kg of supercooled water with a temperature of -5 °C to produce the same amount of ice at that temperature?